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Exogenous crimes and the assessment of public safety efficiency and effectiveness

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Abstract
This work discusses the issue on how to include data about property and violent crimes in the production technology for the assessment of police technical efficiency. It applies recent advances in Directional Distances and Nonparametric Estimators. We claim that crime is an external variable not under the control of the decision units in view of the fact that it is exogenously determined. The results from the Conditional Directional Distance Analysis can be relevant to cities with high property misdemeanors and homicide rates. Our analysis may be helpful to obtain a more robust and fair classification of police and justice units under similar circumstances, determine the empirical effect of crime on police productivity, their optimal input–output relationship, explore potential associations and compensation effects, and rewarding efficient policy makers in the prevention of crime based on measures of police efficiency and effectiveness.

Keywords Data envelopment analysis (DEA) · Directional distance functions (DDF) · Free disposal hull (FDH) · Exogenous factors · Crime · Conditional frontier · Police efficiency · Policing · Pernambuco · Brazil

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1 Introduction

Stimulating best practices and increasing the performance of sworn officers to cope with the critical issue of public safety are some of the main objectives of managing public resources to provide a safe atmosphere for the citizens, and some of the important prospects the Efficiency and Productivity Analysis (PEA) methodologies can provide. The measure of efficiency traditionally derived from PEA methodologies can be understood from the frontier analysis perspective as a ratio of produced outputs over the used resources or inputs. This concept developed by Farrell (1957) and Shephard (1970) was popularized by Charnes et al. (1978). In the so-called Data Envelopment Analysis (DEA) an inefficient unit can become efficient producing more results with the same level of resources or producing the same level of results with fewer resources, or a combination of both. Today, the field of efficiency analysis are wide popularized by means of outnumbered theoretical contributions, surveys, empirical applications and computational tools (Daraio et al. 2019a, 2020; Emrouznejad and Yang 2018; Liu et al. 2013).

Nonparametric Robust Estimators (Daraio and Simar 2005, 2007a, b) are part of the PEA family of frontier estimators proposed to overcome some drawbacks in the traditional DEA and Free Disposal Hull (FDH) measures for the technical efficiency (see discussion provided by Daraio and Simar 2007a). In special, these estimators extend the nonparametric input/output production technology to robustly account for extreme values or outliers in the data, and allow measuring the effect of external environmental variables on the efficiency of Decision Making Units (DMUs). Crime, in our view, is one of these uncontrollable variables that affect the efficiency of many production systems such as public administrations, service units, courts and, in special, police departments. This is because it is strictly related to their performance. Nepomuceno and Costa (2019a, b) and Nepomuceno et al. (2017a, b) discuss the importance of a robust analysis on disaggregated data of property crimes, violent behavior and corruption have on the social prospects for families and institutions. Crime, especially in the Latin American city where this study has been conducted, provokes negative externalities that overflow to the performance of many sectors of the economic activity, and social relations not directly related to their occurrences.

Despite the relevance of this issue, the assessment of public safety performance is one of the least debated issues in the field of Productivity and Efficiency Analysis. In a recent paper discussing the theoretical and empirical advances in productive Nepomuceno et al. (2017) only 22 empirical applications on the police productivity were evidenced in the Web of Science database, having no core publication attached (with high incoming citations). Another recent review on surveys of frontier applications by Daraio et al. (2020) reveals the most prominent areas for empirical applications given by the low amount of published surveys in the specific field and co-occurrence with limited areas of the economic activity. The area of Public Administration and Police, according to the authors, represents a relevant gap that deserves to be filled by new and original empirical applications in the field.

Nevertheless, the literature highlights some important police efficiency analysis. The seminal work in the field can be attributed to Thanassoulis (1995) who assesses the efficiency of 41 British police forces in England and Wales using Data Envelopment Analysis (DEA) and Correlation Analysis. Resorting to labor (officers) and to the occurrences of violent crimes, burglaries, and other crimes as input and to their clear ups rates (the ratio of solved cases by the amount of crime) as output, the author provides important insights for the Parliament Audit Commission focusing on the police procedures and efficiency interrelations. Carrington et al. (1997) studied the efficiency of 163 police units in the Australian state of New South Wales, testing the influence of some socio-economic variables on the scores through regression models. In addition, Diez-Ticio and Mancebon (2002) studied the performance of the Spanish police in solving criminal incidents, and Drake and Simper (2000, 2003) are two similar studies about English and Welsh police forces. In Brazil, some relevant assessments are found in Melo et al. (2005) who measured the efficiency of integrated safety areas (AISs) in the State of Rio de Janeiro using DEA and Scalco et al. (2012) for the efficiency of the Military Police in the state of Minas Gerais.

All those assessments emphasize crime as a resource (input) for the production process or as a product (output) from the production process of police departments. According to Nyhan and Martin (1999) the use of crime data is *controversial* because they are affected by many external factors on which the service unit has no control. The main contribution of this work is to explore a methodology that considers this issue. For this purpose, we contribute to the discussion on crime and punishment by designing a general framework of property crime determination, adapted from the Ostrom (1973) Model for the Security of the Community, which can be expanded to violent crimes and behavior. By developing this general framework in the second section of this article, we explain why the measurement for the technical efficiency of decision units should not rely on the consideration of crime as a discretionary input or output.

We carry out the assessment of technical efficiency by applying Conditional Directional Distances (Daraio et al. 2018; Daraio and Simar 2016). We provide an alternative approach for the assessment of police departments by considering criminal data as *exogenous variables* not under the control of the decision units. We estimate an input-oriented model containing three dimensions of police production in the Brazilian state of Pernambuco; a dimension for Intentional Violent Life-related Crimes (*Crimes Violentos Letais Intencionais*—CVLI), a dimension for the efficiency estimation in one specific property crime: Street Mugging (*Assalto a Transeúnte*) and a last dimension considering Motor Vehicle Robberies (*Roubo de Veículos*). We provide also the assessment of the relative efficiency *conditioned* to the *crimes*, considered as exogenous variables.

Carrington et al. (1997) argues in favor of effectiveness indicators for the performance police services. Because in this field of application there is a clear preference structure on some results more than in others (for instance, in the reduction of homicides prior to street robberies), a efficiency measure and ranking of alternatives may not be sufficient to aid policy makers in the most effective direction. For this concern, we compare our conditional frontier evaluation with an effective measure for the police departments to explore potential associations and insights. This unique evaluation of Pernambuco's Civil Police departments highlights the contribution of conditional measures for the directional efficiency and compares the results with traditional FDH measures for the technical efficiency.

This paper is organized as follows. The next section reviews the existing literature and emphasizes the controversial use of criminal data as input or output in the production process. Section 3 presents the methodology that we propose to include crime in the analysis as an external (exogenous) variable and section. The fourth section describes the data on Pernambuco's municipalities while Sect. 5 is dedicated to the results. It reports the comparative assessment of efficiency models (with and without crime as exogenous determinant), presenting the empirical effects of crime on the technical efficiency of Pernambuco's municipalities. We analyze and compare robust rankings of police departments based on the conditional efficiency and effectiveness. Lastly, the conclusion summarizes the main results and concludes the paper.

2 Review of the literature and the controversial use of crime in the efficiency analyses

Using crime as an input has been reported in many assessments of police efficiency. The seminal application of Thanassoulis (1995) in the British police is the main instance. By using the occurrences of violent crimes, burglaries and other crimes as input to produce clear-up levels on these variables the author emphasizes potential weaknesses in the Police Forces, efficient peers and suggests higher labor associated with higher clear up efficiencies. Another empirical application following this avenue is provided by Verma and Gavirneni (2006) measuring the police efficiency in India using the number of investigated crimes as input on the assumption that it forms the basis for subsequent investigation and prosecution. Property misdemeanors such as burglaries, thefts, robberies, raids, street mugging, vehicle robberies, and violent life-related crimes are typical inputs found in many other performance evaluations e.g. Sun (2002), Barros (2007), Agdokan (2012), Drake and Simper (2000, 2005) and Hadad et al. (2015).

According to Leigh et al. (2019) methods of improving resource efficiency are crucial to public safety. Nevertheless, despite the considerable relevance of input efficiency assessments in the Operational Research literature, we argue crime should not be considered as a discretionary resource because it cannot be directly controlled (reduced) by the police units. According to Gaver and Lehoczky (1987) crime is a stochastic process associated with a function of probability evolving over time with the existing dynamics of arrests and incarcerations. Many social, demographic, spatial and economic covariates not under the control of police stations work as positive determinants for criminality.

For this same reason, it should not be considered as output from the production process. The Carrington et al. (1997) assessment of the Australian Police Patrol in the New South Wales considering property offenses and car accidents as the output to measure the technical efficiency provides one of the many relevant applications in the field. Another interesting application comes from DeAngelo et al. (2014) who considered the inverse of the violent crime rate per capita and property crime per capita as outputs produced by 50 municipal police departments in New York to infer some factors impacting the police productivity (community policing officers, employment, computers).

Similar production setups can be observed with the selection of variables by García-Sánchez (2008) concerning the number of accident reports, Rahimi et al. (2017) considering car crashes and many other assessments such as Nyhan and Martin (1999), García-Sánchez (2007), Barros and Alves (2005), Gorman and Ruggiero (2008), Wu et al. (2010). Table 1 summarizes the production structure configuration for some selected assessments of police performance using the crime data as input or output. Because of the stochastic nature of the criminal incidence, it is always possible to assume an increase in the police technical efficiency (e.g. by increasing the number of solved cases) and still have a more than proportional increase in the criminal incidence regardless how productive the police is. Thus, efficiency analysis of police units can never rely on an investigation of crime as output (Ostrom 1973).

The criminal occurrences for a specific region or period are influenced by exogenous determinants, which (in the short run) cannot be controlled directly or are difficult to manage. The Nobel laureate Professor Elinor Ostrom (1973) proposed a general framework for the propensity of individuals toward unruled strategies that jeopardize the security of the community. An adaptation for this model on property offenses is illustrated in Fig. 1. According to Ostrom, population characteristics such as income, social status, educational

Table 1 Selected literature on police efficiency assessments

Publication	DMUs	Inputs	Outputs
Thanassoulis (1995)	41 British police forces in England and Wales	Police Personal (officers); Violent Crimes; burglaries; Other crimes	Violent Crimes Clear-ups; Burglaries Clear-ups; Other Crimes Clear-ups
Carrington et al. (1997)	163 Patrol units in New South Wales (Australia)	Police Officers, Civilian Employees; Capital Equipment	Offences; Arrests; Summons; Car accidents; Kilometers travelled by police cars
Gorman and Ruggiero (2008)	49 US states	Police officers, Other employees; Vehicles	Murders; Other Violent Crimes; Property Crimes
Nyhan and Martin (1999)	25 Local Governments (US)	Police staff (FTE); Department cost	Violent and Property Crimes Clear-ups; Response Time; Crime Rate
Drake and Simper (2000)	35 British police forces in England and Wales	Employment cost Premises-related expenses Transport-related expenses Capital and other expenses	Traffic offenses; Crime Clear-ups
Drake and Simper (2005)	41 British police forces in England and Wales	Burglaries; Motor Vehicle Robberies; Budget	Civilian days lost; Aggregate Crimes Clear-ups
Verma and Gavirneni (2006)	25 states (India)	Expenditure; Police officers; Investigating officers; Investigated Crimes	Persons arrested; Persons charge sheeted; Persons convicted; Trials completed
Barros (2007)	33 Lisbon police precincts over 3 years (99 observations)	Police officers; Cost of labor; Vehicles per precinct; Other costs; Thefts; Number of burglaries; Car robberies; Drug-related Crimes	Theft Clear-ups; Burglaries Clear-ups; Stolen cars Clear-ups; Drug-related crimes Clear-ups; Police raids; Number of stop operations; Minor offenses
Aristovnik et al. (2014)	76 Police organizations in Slovenia	Posts occupied; Work stations; Police vehicle radio stations; Criminal offences[a]; Violations of public order regulations[a]; Road accidents[a]	Crime Clear-ups; Road accidents (serious injury and minor injury); Average response time of police patrols; Usage of instruments of restraint and warning shots
Hadad et al. (2015)	13 police stations in the south of Israel over 4 years (52 observations)	Property crimes; Violent crimes; Burglaries; Traffic accidents; Traffic accidents with injuries. Personnel and operational costs; Population size; Vehicles	Property crimes Clear-up rates; Drunken driving cases exposed; Traffic reports

[a]Non-discretionary inputs

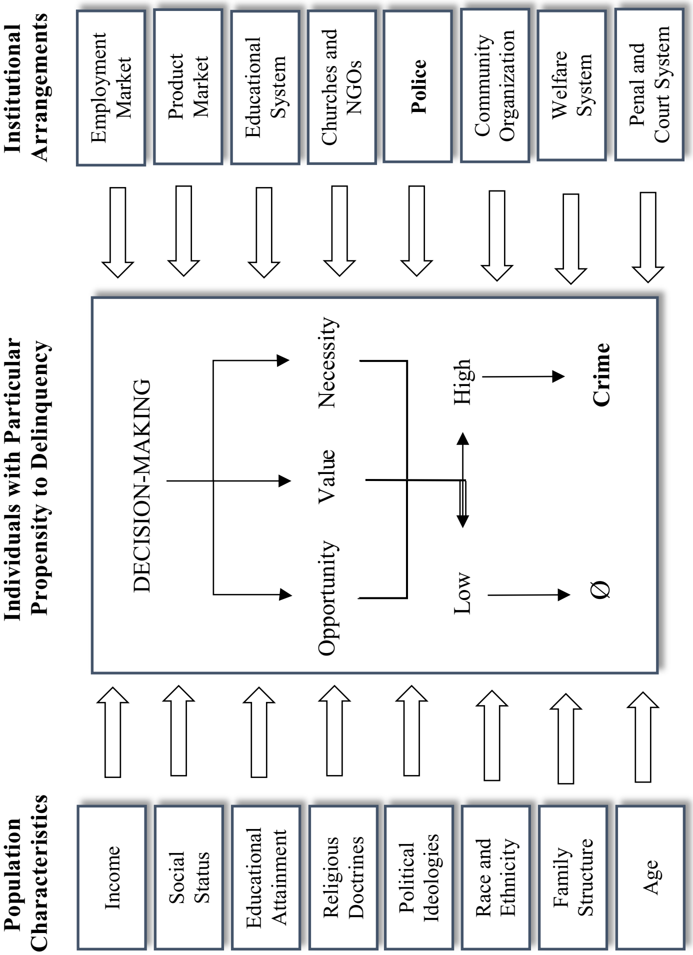

Fig. 1 General framework of crime determination. Adapted from Ostrom (1973) Output Model for the Security of the Community

attainment, religious preference, density, social status, age, home ownership, race and ethnicity are social-economic and demographic indicators of how well institutional arrangements such as the employment, product, housing market, educational, court, penal and welfare systems and the community organization have operated in the past (long-run). The lower the level on those indicators, the higher the propensity of individuals within the community to pursue criminality.

The decision-making toward crime described by the framework in the Fig. 1 can be conceptualized as social interactions mediated by multiple institutional arrangements that are not under the control of the police unit. Notwithstanding, the police is one of these many arrangements. The individuals with sufficient propensity to delinquency are represented in the middle of this framework. In this case, there are three main triggers for a prone individual to get involved in the criminal conduct: the opportunity, the expected value, and the necessity. Only one of these triggers is sufficient to ignite the misdemeanor, given the operating circumstances. A distracted passerby, the lack of policing or the lack of appropriate attention on personal belongings can be characterized as opportunities for a prone individual to engage in property misdemeanors when they do not present high expected value or necessity for this strategy. The expected value for the crime is defined under the rational choice perspective of the economist and Nobel laureate Gary Becker (1968). The prone individual's decision to pursue the offense is made when the expected utility from engaging in the criminal activity is positive, i.e. the expected gains are sufficiently high to compensate the expected loss (in terms of the probability of punishment). Likewise, in this case the decision toward crime can be made even when the necessity or opportunity is not favorable.

The last trigger for criminality in prone individuals refers to the necessity to subtract the property of others. Critical poverty scenarios encourage a rational individual to irrational decisions, such as to commit the crime when the probability of being caught, exposed and convicted is high (low expected value) or when the circumstances are not favorable (low opportunity). While only one trigger is sufficient to ignite the desire toward the property offense, given the operating circumstances, low levels in all of them are required for a prone individual to restrain from committing the offense. Naturally, many individuals who despite the opportunity, the expected value and the necessity, will not lean upon delinquency. In the Ostrom (1973) perspective, those are individuals with considerable levels of education attainment, higher age, status, income, religious and political doctrines. Thus, cannot be characterized as individuals prone to crime in the decision model of Fig. 1.

These listed population covariates, besides not being under the control of police units, they cannot be sufficiently enhanced or coherently measured in the short-run, making it impossible to predict a reduction in the criminal input or an expansion in the security of the community. In Brazil, one of the most well deemed state-level programs in the prevention of homicides is the so-called *Pacto pela Vida* (A Pact for Life), executed in many northern regions of the country (Ratton et al. 2014; Ratton and Daudelin 2018). Today is of common knowledge from empirical evidence that some of the good prospects in reducing the criminal behavior can be attributed to the country's long-term favorable economic stability. Despite the increasing efforts from the recent administrations, crime has come to increase with the increase in the unemployment rate.

If it were possible to trace every single aspect and determinants in modelling the criminal behavior, an efficiency measure for the police performance, using crime as discretionary input or output would still be potentially biased by the correlation and dependence on other results. Asmild et al. (2011) using Pastor et al. (2002) statistical test for nested radial frontier models exemplifies this assertion. The authors divided a sample with 16 police units in three main production functions in order to investigate the police performance: an enforcement model, a response to emergency model and a crime prevention model. Each model shares the same resource (labor in full time equivalent measure) to produce different results. After performing further analysis for the third model, the authors opt to remove the inverse crime rate from the assessment for being redundant in view of the fact that this variable can be returned as a consequence from the other two considered outputs (time on patrol and the number of times people are stopped and interviewed).

Another important question concerns the simultaneity bias between the police activity and the occurrence of crimes. The expected effect of arresting criminals in a location is to reduce the occurrence of crimes. However, it is only possible to arrest criminals because there was crime. Simultaneity bias causes a spurious correlation between inputs and outputs in the model. Gordon (2010) discusses this bias and other problems related to criminal variables. In this way, it is essential that crime be considered an external variable in the efficiency analysis of police performance.

The problem with efficiency assessments based on criminal data as input and output is also acknowledged by some interesting applications such as Diez-Ticio and Mancebon (2002) investigating 47 Spanish Police Forces using officers and vehicles as inputs and clear-ups as outputs; Domínguez et al. (2015) using panel data on police efficiency and exogenous probabilities; Aristovnik et al. (2013) considering previous studies on the role of policing in their specifications of inputs and output; and Barros (2007) who discuss the subject with data provided by the *Policía Nacional Civil* in Guatemala. The last author suggests two potential solutions to this problem. The first is the exclusion of the external crimes outside the control of police management in order to decrease the degree of bias in the inefficiency distribution. The second is to include the uncontrollable criminality and investigate the significance of the bias by comparing efficiencies scores with and without the external factors. As acknowledged by the author, removing important data from the analysis limits the potential prospects from the results. In addition, including environmental uncontrollable factors as input does not guarantee the consistency of the results. Nevertheless, the author opts for the exclusion of uncontrollable inputs from the efficiency analysis.

An alternative to the exclusion of crime data from the analysis is considered by Aristovnik et al. (2014) in the assessment of 76 Police organizations in Slovenia. The authors use as inputs the number of posts occupied, work stations, police vehicle radio stations and include the occurrences of criminal offenses, violations of public order regulations and road accidents as non-discretionary inputs in the production technology. By restricting the reference set and the variables that are active in the decomposition of the technical efficiency measures, or by imposing a directional vector equal to 0 (zero), non-discretionary models and models for undesirable data are able of controlling for fixed resources that cannot be reduced or negative outputs that should not be increased, such as externalities.

As highlighted by Verma and Gavirneni (2006), one of the fundamental aims for the police enforcement is to prevent crimes, but it is impossible to measure the number of crimes that have been prevented, only those that have been solved. The next section presents an alternative. This discussion is important to highlight the essential argument for crime to be considered as an external factor or as a non-discretionary variable in the efficiency analysis of police performance at the short-run. Strong public policies such as strict jail sentences, incentives for social inclusion, income and educational attainments, can provide a disincentive to committing crime. Still, it is extremely difficult to attribute these public policies to the performance of a specific police station, and it is difficult to perceive the effects of these policies at the short-run reduction of crime

The proposed methodology has the additional benefits of identifying the influence of exogenous factors, such as crime, in the distribution of the inefficiency, and thereby determine whether they can be considered as non-discretionary inputs or outputs in the efficiency analysis. For both orientations, conditional directional distances are proposed as robust measures for the technical efficiency of police departments that considers the exogenous effects of crime, neither assuming it as input nor output, rather an environmental factor not under the control of the policymaker at the short-run.

3 Methodology

3.1 Unconditional directional efficiency

The traditional methods to evaluate the productive performance of Decision Making Units (Farrell 1957; Charnes et al. 1978) involve measuring the technical efficiency as a radial projection from a specific production configuration containing $i=1, 2, 3 \ldots, n$ inputs and $r=1, 2, 3 \ldots, s$ outputs for a given DMU j from m decision units to an efficient frontier Ψ^* in the production set $\Psi_{(x,y)} \in \mathbb{R}^{n+s}$ so that x can produce y. An adaptation for this traditional evaluation consider time-series data for a given DMU j in the moment t from p periods to estimate the internal or inverse production technology (Nepomuceno and Costa 2019c; Nepomuceno et al. 2020a, b). In the mathematical formulations, the efficient frontier is composed by extreme production observations that "envelop" the remaining set, and the efficiency score can be estimated in either input contractions (keeping the current output levels) or output expansions (keeping the current input levels):

$$\Psi^* = \{(x, y) \in \Psi_{(x,y)} \Big| \left(x\lambda^{-1}, y\lambda\right) \notin \Psi \forall \lambda > 1 \tag{3.1}$$

where Ψ is the set of feasible combinations of inputs and outputs of the production technology.

The Directional Distances Functions proposed by Chambers et al. (1996, 1998) and Färe and Grosskopf (2004) to measure the performance of decision units are a more flexible nonparametric methodology which generalize radial projection to gauge the efficiency. The efficiency is assessed by choosing a feasible direction $g = \mathrm{g}_{(x,y)} \geq 0$ where the resources (inputs) must be reduced and/or the outputs expanded in order to reach the efficient frontier:

$$\beta(x, y|g_{(x,y)}) = \sup\left\{\beta > 0 | \left(x - \beta g_{(x)}, y + \beta g_{(y)}\right) \in \Psi\right\} \tag{3.2}$$

The directional input vector must attain to contract the production resources whereas the directional output vector must attain to expand the production products. From an economic perspective, this is consistent with a maximizing behavior of the analyzed units (Färe and Grosskopf 2004; Färe et al. 2015). The unit vector $\mathrm{g}_{(x,y)}=1$ was chosen as the feasible direction to simplify the aggregation of (in)efficiency scores (Färe et al. 2015). The choice for the unit vector $\mathrm{g}_{(x,y)}$ is the most common and straightforward because units are evaluated equal-proportionally in the same direction. Consider an input directional distance function:

$$\beta(x, y|g_{(x)}) = \sup\left\{\beta > 0 | \left(x - \beta g_{(x)}\right) \in \Psi\right\} \tag{3.3}$$

Consider X the minimum usage of resources coherent with a minimizing behavior. Then, the input directional distance function turns to:

$$\begin{aligned} \beta\left(x, y | g_{(x}\right) &= \sup\left\{\beta > 0 | \left(x - \beta g_{(x)}\right) \geq X \in \Psi\right\} \\ \beta\left(x, y | g_{(x}\right) &= \sup\left\{\beta > 0 | \left(x - \beta g_{(x)}\right) \geq X\right\} \\ \beta\left(x, y | \mathrm{g}_{(x}\right) &= \frac{x - X}{\mathrm{g}_{(x)}} \end{aligned} \tag{3.4}$$

This value β turns out to be an inefficiency score as it is the difference between the observed (x) and the minimum (X) input when we opt for the unit vector ($\mathrm{g}_{(x)} = 1$). In this case, the aggregate (industry) distance function is simplified to the sum of the individual service units distance functions (see Färe et al. 2015 for the analogous output distance function case). In practice, this can be interpreted as having a policymaker's preference equally distributed on all the production resources, which guarantees one not to excel in importance to the reduction of the other.

Although the unit vector $\mathrm{g}_{(x,y)}$ is the most common and simple choice to determine an absolute measure for the technical inefficiency for each decision unit, other possibilities are viable. Wang et al. (2019) provide an interesting and comprehensive review on the existing methodologies to project the inefficient decision units onto the efficient frontier. Other discussions are provided by Fukuyama and Weber (2009), Fukuyama and Weber (2017), Peyrache and Daraio (2012), Daraio and Simar (2016) and Nepomuceno et al. (2020a, b, c). In this framework, if units have $\beta = 0$ they are considered efficient benchmarks at the boundary of their production frontier. Directional Distances offer some advantages:

- Directional Distances provide a non-radial projection for the technical efficiency measured by the directional vector $g = \mathrm{g}_{(x,y)}$ useful to many production sets specified by multiple input–output orientations for $\beta(x, y)$ to be optimized such that $x - \beta\mathrm{g}_{(x)}, y + \beta\mathrm{g}_{(y)}\big) \in \Psi$ (Färe and Grosskopf 2004; Färe et al. 2008a, b);
- The additive nature of the Directional Efficiency permits the inclusion of negative (undesirable) inputs and/or outputs in the production technology, supporting complete assessments in many industrial and service units (Daraio et al. 2018; Portela et al. 2004; Simar et al. 2012);
- Flexibility for the decision maker to pursue different strategies in the direction specification to project inefficient units to the efficient frontier. Instances are data-driven approaches (Daraio and Simar 2014, 2016), lacks and slacks projections as a proportion of the resources and results of a reference unit (Fukuyama and Weber 2017; Nepomuceno and Costa 2019c), inclusion of subjective (non-compensatory) preferences for the reduction of some inputs more than others (Nepomuceno et al. 2020a, b, c).

The attainable efficient frontier in the Free Disposal Hull (FDH), which assumes only the free disposability of inputs and outputs, is usually obtained from Linear Programming (LP) envelopments of the observed units (Deprins et al. 1984). Simar and Vanhems (2012) extended the probabilistic approach by Daraio and Simar (2005, 2007a) to the directional distances introduced below:

$$\beta\left(x, y | \mathrm{g}_{(x,y)}\right) = \sup\left\{\beta > 0 \middle| H_{(XY)}\left(x - \beta\mathrm{g}_{(x)}, y + \beta\mathrm{g}_{(y)}\right) \in \mathbb{R}^{n+s} > 0\right\} \tag{3.5}$$

where $H_{(XY)}(x, y) = prob(X \leq x, Y \geq y)$ is the joint probability that a given unit j presenting the production configuration (X, Y) dominates the point under evaluation with production configuration (x, y).

The Free Disposal Hull (FDH) measures for the technical efficiency is interesting for assuming only the free disposability of inputs and outputs, which makes the analysis robust to the convexity assumption implied by DEA studies. Indeed considering that also a linear combination of existing police offices is feasible is quite a restrictive assumption for our context and for this reason we decided to avoid it in our analysis. The variables and the model formulated in this paper can be estimated with a DEA approach provided that we are able to justify the convexity assumption or to test for it. These further analyses are left for future developments.

3.2 Conditional efficiency

Exogenous factors cannot be considered neither as an input nor as an output of the production process. Nevertheless, they affect the performance of the firms and, at some instances, can explain the distribution of efficiency scores, spillovers, economies of scale, scope, measures for quality and effectiveness and the shape and potential shifts in the production technology over the time. Especially considering effective units, they must be capable of handling non-discretionary conflicting outcome measures in terms of environmental and exogenous-determined factors to produce desired outcomes (Lewin and Minton 1986).

To this topic has been given attention since the seminal model of Banker and Morey (1986) considering non-discretionary exogenously defined inputs and outputs in the estimation of the efficient frontier for a network of fast food restaurants. Ruggiero (1998) adapts prior modelling (Ruggiero 1996) for measurement of the technical efficiency with non-discretionary inputs to cope with the overestimation of technical inefficiency in Banker and Morey (1986) approach by restricting to zero weights with higher levels of non-discretionary resources. Banker and Morey (1986) also serves as base for the generalized DEA formulation of Syrjänen (2004) and many other approaches (examples can be found in Camanho et al. 2009; Cordero-Ferrera et al. 2010; Karagiannis 2015; Ray 1991; Simar and Wilson 2007).

Exogenous factors can affect the production process by either changing the (X, Y) configuration on the attainable set $\Psi^* \in \Psi_{(x,y)}$, shifting the efficient boundary, or by affecting the distribution of inefficiencies, and the probability for the unit under evaluation to reach its optimal input/output ratio. It is possible that both effects are present or completely independent of the production process. Alda (2014) and Alda and Dammert (2019) provide an interesting perspective on this. The authors, based on postulates of the contingency theory, argue that because of the public nature of police organizations' activities, they are heavily influenced by the environmental factors which invariably affects their efficiency.

Some procedures to consider the role of external factors have been proposed. Most of them are based on two or multiple stages and regression inferences, which, as discussed by Simar and Wilson (1998, 2007) and Daraio and Simar (2007a) are flawed by the development of biased estimators of the efficiency scores, serially correlated estimates and by the necessity to impose parametric assumptions on the production process. For a state-of-the-art literature review on the inclusion of environmental variables in nonparametric frontier models, see Bădin et al. (2014).

Conditional efficiency measures, accounting for environmental uncontrollable factors (Z), were introduced by Daraio and Simar (2005). They are based on the basic idea that the joint probability distribution $H_{(XY)}(x, y) = prob(X \leq x, Y \geq y)$ when conditional on $Z=z$ can define an attainable production set $\Psi^z = (x, y) \in \mathbb{R}^{n+s}$ such that x can produce y if $Z=z$. For the directional efficiency, the conditional directional distance function is defined as (Daraio and Simar 2014, 2018):

$$\beta\left(x, y|z, g_{(x,y)}\right) = \sup\left\{\beta > 0|H_{(XY|Z)}(x - \beta g_{(x)}, y + \beta g_{(y)}|z) \in \mathbb{R}^{n+s} > 0\right\} \tag{3.6}$$

where $H_{(XY|Z)}(x, y|z) = prob(X \leq x, Y \geq y|Z = z)$ is the joint conditional probability that a given unit j presenting the production configuration (X, Y) dominates the point under evaluation with production configuration (x, y). For inactive cases (those with $g = \mathrm{g}_{(x,y)}$ input/output directions = 0):

$$\beta\left(x, y|z, g_{(x,y)}\right) = \sup\left\{\beta > 0|H_{(X_1Y_1|X_2Y_2Z)}\left(x_1 - \beta g_{(x_1)}, y_1 + \beta g_{(y_1)}|x_2, y_2, z\right) \in \mathbb{R}^{n+s} > 0\right\} \tag{3.7}$$

where $H_{(X_1Y_1|X_2Y_2Z)}\left(x_1, y_1|x_2, y_2, z\right) = prob\left(X_1 \leq x_1, Y_1 \geq y_1|X_1 \leq x_1, Y_1 \geq y_1, Z = z\right)$ is the joint conditional probability that a given unit j presenting active production configuration (X_1Y_1) and inactive production configuration (X_2Y_2) dominates the point under evaluation with production configuration (x, y). Such measures are important for the distance function flexibility in the case of non-discretionary resources and/outcomes.

The joint conditional probability measure $H_{(XY|Z)}(x, y|z) = prob(X \leq x, Y \geq y|Z = z)$ is empirically estimated by a non-parametric estimator that smooths the multiple $Z_{(c)} \in \mathbb{R}^k|c = 1, 2, 3, \ldots, k$ values in the neighborhood of z from a sample $S_{(p)} \in \mathbb{R}^{n+s}$ of $q = 1, 2, 3, \ldots, p$ observations:

$$\hat{H}_{(XY|Z)}(x, y|\mathrm{Z} = z) = \frac{\sum_{q=1}^{p}\begin{cases}(X_q \leq x, Y_q \geq y)K_h\left(Z_q, z\right) \Leftrightarrow X_q \leq x, Y_q \geq y\\ 0, \quad otherwise\end{cases}}{\sum_{q=1}^{p} K_h\left(Z_q, z\right)} \tag{3.8}$$

where $K_h\left(Z_q, z\right)$ is the appropriate kernel estimator and h the bandwidths vector. As discussed by Daraio and Simar (2007a, b) the estimation of conditional full frontiers does not depend on the chosen kernel, although a kernel with compact support is recommended. Choices for a Gaussian kernel in this non-parametric setup makes it difficult to detect any influence of the exogenous factors. The conditional measure however, is sensitive to the selected bandwidths (see Bădin et al. 2010, 2012, 2014, 2019).

4 Data

The data set provided by the *Secretaria de Defesa Social do Estado de Pernambuco* (Pernambuco's Secretariat for Social Defense) corresponds to the number of police officers (*Efetivo Policial*), the occurrences of homicides and other violent life-related offenses (CVLI—*Crimes Violentos Letais e Intencionais*), street mugging (*assaltos a transeunte*) and vehicle robberies/thefts (*roubo de veículos*), and the data on their clear-ups for the year 2017, considering 145 of the 185 Pernambuco municipalities (cities). A small part of the clear-up data corresponds to outputs from crimes committed years before. Some crimes such as murder take longer to be solved with the identification of the offender (beyond the year period) and as such, these output measures are biased. Nevertheless, regarding a dataset having 24,613 criminal records, we expect these specifics input-to-output configurations beyond the time lapse does not affect the overall aggregated evaluation. Table 2 describes some descriptive statistics (range, quartiles, median, mean, and standard deviation) for the corresponding variables.

As input, only the number of police officers (civil servants) is considered. This is the most recurrent and relevant input considered for the investigation of the technical efficiency of police units in many important assessments such as Thanassoulis (1995), Carrington et al. (1997), Gorman and Ruggiero (2008), Nyhan and Martin (1999), Drake and Simper (2000), Verma and Gavirneni (2006), Barros (2007) and Aristovnik et al. (2014). Other commonly used resources are patrol vehicles, stations, surveillance cameras, personal and operation costs and other related expenditures. Because of some data limitations, only human resource are considered. There is an increasing effort by the universities and research centers with the Secretaria de Defesa Social (Pernambuco's state secretariat for public security) in the formalization of institutional agreements for the creation of databases to track internal resources and exogenous criminal records, and for disclosure of data. This limitation is expected to be minimal for future assessments.

The differences between police stations are regarded by considering the criminal records used as exogenous factors to decompose conditional directional measures for police efficiency at the municipal level. For instance, small rural municipalities are expected to have a small number of occurrences compared to urban cities. The inclusion of crimes as exogenous determinants for the police productivity works as control variables for the assessment, keeping the assumption other institutional, social and economic determinants can affect crime preserving the appropriate proportions.

Brazil has headed the ranking of homicides in the world (about 60 thousand per year according to the most recent data from Cerqueira et al. 2018) and Pernambuco has been responsible for about 50% of the increase in the violence during 2017 (Jornal do Commercio 2017). In fact, the state of Pernambuco has experienced both a significant reduction and increase in the violence during the last decade. The Pact for Life program, one of the most successful state-level projects to reduce the occurrences of homicides and other violent crimes, had its contribution by integrating the public policy of security with the Judicial Power, Legislative Power, municipalities, state and federal government. The value added by this integration has been fundamental for the fact police cycle is not unitary, meaning that the police that arrive at the crime scene are not the same one that investigates. Some important studies of crime to understand the context in Pernambuco are Daudelin and Ratton (2017), de Gusmão et al. (2019, 2020), Figueiredo and Mota (2016), Nepomuceno and Costa (2019a), Pereira et al. (2015, 2017), Silva et al. (2017) and Mota et al. (2020).

Table 2 Descriptive statistics

Variables		Min.	25th Qt.	Median	Mean	75th Qt.	Max.	SD
Input	Police Officers	3.000	6.000	8.000	9.862	11.000	48.000	6.36
Outputs	CVLI Clear-ups	0.000	3.000	5.000	8.359	11.000	42.000	9.24
	Street Mugging Clear-ups	0.0	1.0	5.0	9.2	10.0	79.0	13.24
	Vehicle Robberies Clear-ups	0.000	0.000	1.000	2.055	2.000	25.000	3.89
Crimes	CVLI	0.00	7.00	13.00	20.03	26.00	198.00	23.94
	Street Mugging	2.0	19.0	42.0	144.1	135.0	2198.0	300.9
	Vehicle Robberies	1.00	11.00	25.00	70.21	63.00	1161.00	127.92

There are two kinds of Police: the Military Police working on the streets to guarantee public safety, and the Civil Police that investigates the crimes. The assessment provided in the next section refers to the Civil Police production only, without depreciation on the effort of both police to develop integrated actions to reach the objectives. The focus of the Pact for Life program and the past administrations is on the reduction of homicides prior to any other criminal occurrence. This objective has been taken into consideration to provide a ranking of Pernambuco's municipalities based on the effectiveness of the output. Many important cities in the state were removed from this assessment because of the impossibility to work on missing data. Nevertheless, the proposed methodology can be easily extended to consider the aggregate scenario and prospects for the public safety policy.

5 Results

The first part of this section compares conditional and unconditional efficiency measures of police productivity. In the second part we compare Efficiency with Effectiveness. We adopt an input-oriented scenario for practical reasons; assuming reductions in the police personal are always feasible. Given the diverse complexity in many criminal reports, at a micro level is easy to assume that an increase in the police efforts may not necessarily guarantee an increase in the clear up rates. For this reason, the concept of efficiency in the evaluations are defined as the feasible contraction in the resources keeping the regularity in the outcome.

Effectiveness can be defined as the ability of producing a desired or intended output level. It requires predefined values and judgments over specific attainable goals. Police departments can be efficient in the usage of resources to produce public safety yet not effective in producing some of the more desirable results. In the case of Pernambuco, the *Pacto pela Vida* has established the main values guiding the construction of the public security policy plan with the prioritization of reducing intentional lethal violent crimes (CVLI) by 12% a year in Pernambuco. This specific goal is taken into consideration to classify the units according to an efficacy measure and investigate a potential relationship with the directional efficiency conditional to the exogenous criminality. Following Thanassoulis (1995), cross-correlation statistics are provided in order to explore potential compensation effects.

5.1 Comparing conditional and unconditional efficiency

We investigate the effect of crime incidence on the performance of police departments to the decomposition of absolute measures for the Directional FDH technical efficiency, and the construction of more robust parallel efficiency rankings conditional to the external factors. The choice for the bandwidth in the conditional case was made according to Badin et al. (2019). The direction was defined by the unit vector: $\mathrm{D}_t(\mathrm{gx}, \mathrm{gy}) = (1, 1)$ for both the conditional and unconditional FDH directional measures for the technical efficiency, which implies all observations evaluated in the same direction (Färe and Grosskopf 2004).

First, we want to investigate how crime may influence the efficiency of policing. The local analysis proposed by Badin et al. (2012), permits to regress the local effect of individual radial projections of the production technology conditional to each operating factor Z on the shift of the reachable frontier so that the resulting influence can be observed by means of the frontier ratio (unconditional $\hat{\beta}(x, y)$ and conditional $\hat{\beta}(x, y|z)$ measures). Each exogenous crime may correlate with each decision unit production set in a particular trajectory beneficial or detrimental to the current performance of the decision unit. The following Figs. 2, 3 and 4 present the scatter plots of the ratio $Q_z = \frac{\hat{\beta}(x,y|z)}{\hat{\beta}(x,y)}$ on each Z exogenous determinants of the police productivity (respectively CVLI, Street Mugging and Vehicle

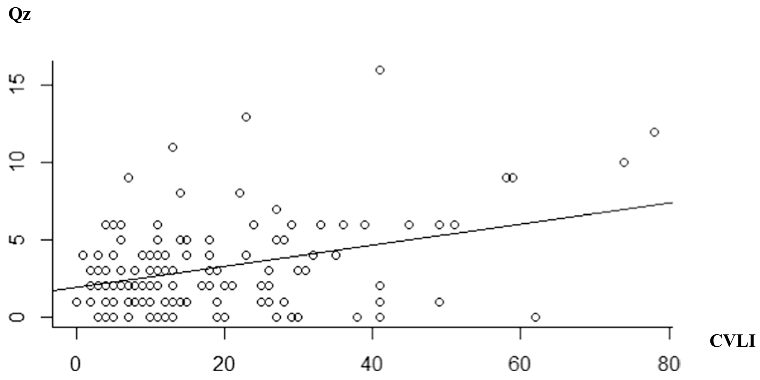

Fig. 2 Empirical effect of homicides (CVLI) on the technical efficiency of policing

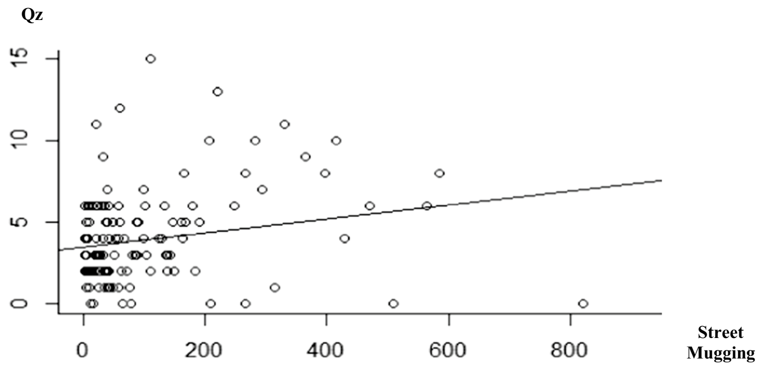

Fig. 3 Empirical effect of street mugging on the technical efficiency of policing

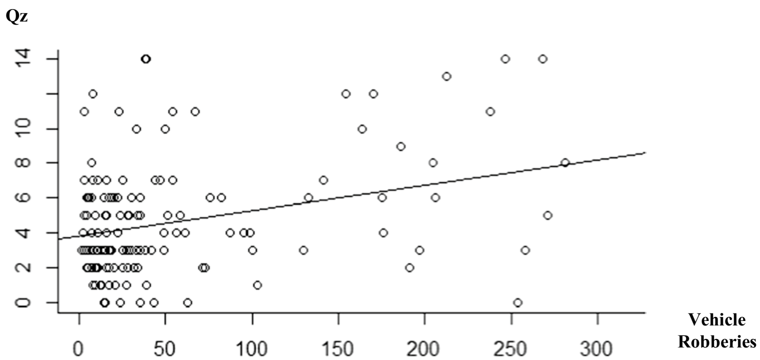

Fig. 4 Empirical effect of motor vehicle robberies on the technical efficiency of policing

Robberies). In an input oriented framework as the one adopted here, a decreasing nonparametric line shows that the considered external factor is favorable to efficiency improvement. Increasing nonparametric regression lines means that the considered exogenous factor is unfavorable to efficiency improvements.

The plots in Figs. 2, 3 and 4 do not report the four municipalities with the highest occurrences of Violent Crimes (CVLI), Street Mugging and Vehicle Robberies which are the municipalities of Cabo de Santo Agostinho, Camaragibe, Santa Cruz do Capibaribe and Vitória de Santo Antão. The empirical effect of crime can be better identified with the removal of these outliers from the local analysis presented.

All the three types of crime present light positive slopes, suggesting the presence of light unfavourable effects on the median frontier level. This means that the probability of being far from the frontier (being less efficient) might be increasing for municipalities with higher levels of these crimes (z values). Thus, when conditional to these exogenous determinants, it is plausible to assume that some inefficient units may become less inefficient, i.e. the shift and distribution of inefficiency changes considering the crime as exogenous to the production technology. This may well be the case: for most of the decision units, the bigger the number of criminal occurrences the higher the probability of more severe crimes to come and become a priority in the police department. Those crimes usually take more time and resources than less severe crimes, and so have a negative impact on the efficiency of the police departments, but not necessarily in the efficacy of policing that, as we may see, is related to a predefined policy preference on some input/output ratio.

The development of unconditional and conditional input-oriented FDH directional efficiency measures for the 145 police departments in Pernambuco are summarized as percentages distributions in Table 3. The entire set of absolute and relative inefficiency measures are provided in Table 10: Results from the Directional Efficiency Evaluation in the "Appendix" section of this manuscript. The absolute measures for the technical inefficiency are given as police personal slacks such that efficient units score 0 (zero) in the results. The relative measures is obtained dividing the absolute inefficiency by the number of the related input for the specific unit under evaluation.

For all the three dimensions of the police production, the number of efficient benchmarks for the remaining inefficient units is higher when conditional to the crimes as uncontrollable factors than it is considering common directional efficiency measures from the FDH application. In the conditional case, we have 23 units (15.86%), 13 units (8.96%)

Table 3 Percentage distribution of police departments by inefficiency intervals

Relative efficiency	FDH directional efficiency			Conditional FDH directional efficiency		
	η(i) = 0.3333; η(ii) = 0.4444; η(iii) = 0.5000; μ = 0.391; sd = 0.1826; $\sum s^-$(iii) = 724; h^+ = 0.7			η(i) = 0.3333; η(ii) = 0.4285; η(iii) = 0.5000;. μ = 0.3808; sd = 01909; $\sum s^-$(iii) = 693; h^+ = 0.7		
	CVLI	Street mugging	Vehicle robberies	CVLI	Street mugging	Vehicle robberies
0.00 (efficient)	14.48%	6.21%	4.82%	15.86%	8.96%	8.27%
0.01–0.25	24.82%	14.48%	6.90%	24.82%	13.79%	6.90%
0.26–0.50	53.10%	48.27%	49.65%	51.72%	46.90%	47.58%
0.51–0.75	7.59%	31.03%	38.62%	7.59%	30.34%	37.24%
0.76–1.00	–	–	–	–	–	–

and 12 units (8.27%) scoring zero inefficiencies for the CVLI, Street Mugging and Motor Vehicle Robberies production sets, respectively, compared to 21 (14.48%), 9 (6.21%) and 7 (4.82) units in the directional FDH evaluation. In addition, the intervals for the relative inefficiency are different. With exceptions on the proportion of units in the interval 0.01 to 0.25 for the Vehicle Robberies production dimension and the proportion of units in the interval 0.51 to 0.75 for the violent crimes (CVLI), the remaining intervals for the inefficient units present ranking changes with different proportion of units in each dimension.

This is indeed relevant from a practical point of view for the management of resources and recognitions. Table 4 highlights the nature of this consideration comparing the ranking changes in 5 municipalities in the CVLI efficiency evaluation. Timbaúba in this assessment presents the most considerable ranking reversal from the position 16 (0.353 inefficiency) to share the first position with the other 22 regions. The other municipalities present relevant improvements in the efficiency evaluation as well. The most drastic change are observed in the Street Robberies assessment for Moreno, which moves from the position 25 presenting 0.556 inefficiency to the first position and for Cabo de Santo Agostinho, which moves from the position 19 presenting 0.500 inefficiency to the first position presenting 0.000 inefficiency (see "Appendix" section). Other considerable changes can be observed for the municipalities of Vitória de Santo Antão from 13ª position to the fist and Vertentes from the 11ª position to the fourth in the Motor Vehicle Robberies evaluation.

Because the average inefficiencies are smaller for each dimension in the conditional FDH assessment compared to the traditional non-parametric directional distance efficiency measures, the highest sum of slacks to be reduced is smaller (693 compared to 724 in the Motor Vehicle Robberies dimension assessment), despite the small difference between the standard deviations on both type of assessments. This provides a fairer evaluation on the police departments. Besides robust, more municipalities are deemed efficient in the cities conditional to different criminal and social structures. The following Tables 5, 6 and 7 present the efficient units for each considered police production dimension (highlighted municipalities included from the conditional frontier evaluation).

5.2 Comparing efficiency with effectiveness

Many efficient units may not be effective to reach some preferred outcomes. While efficient benchmarks present the right way to reach an objective, effective units present the right objectives to be reached. The predefined goal in reducing 12% of CVLI criminality by the *Pacto pela Vida* program, at the state level, permits further inference on whether efficiency comes with the additional cost of effectiveness for the police units in the conditional

Table 4 Ranking reversals in the CVLI evaluation

Municipality	FDH efficiency rankings		Conditional FDH rankings	
	Position	Efficiency score	Position	Efficiency score
Timbaúba	16ª	(0.353)	1ª	(0.000)
Venturosa	4ª	(0.125)	1ª	(0.000)
Nazaré da Mata	10ª	(0.272)	7ª	(0.182)
João Alfredo	18ª	(0.363)	11ª	(0.273)
Toritama	22ª	(0.428)	17ª	(0.357)

Table 5 Efficient units (CVLI)

AGUAS BELAS
BARREIROS
BELO JARDIM
BONITO
CABO DE SANTO AGOSTINHO
CAMARAGIBE
CAMOCIM DE SAO FELIX
CANHOTINHO
CASINHAS
CUPIRA
GOIANA
IATI
JATAUBA
JUCATI
LAGOA DO CARRO
PARANATAMA
SALOA
SAO BENEDITO DO SUL
SAO JOAQUIM DO MONTE
TAQUARITINGA DO NORTE
TIMBAUBA
VENTUROSA
VITORIA DE SANTO ANTAO

The units in bold are conditionally efficient, i.e. they become efficient taking into account the exogenous variable represented by number of crimes

Table 6 Efficient units (street mugging)

ALTINHO
BELO JARDIM
CABO DE SANTO AGOSTINHO
CAMOCIM DE SAO FELIX
CARPINA
JUCATI
JUPI
LAJEDO
LIMOEIRO
MORENO
PESQUEIRA
SALOA
SAO LOURENCO DA MATA

The units in bold are conditionally efficient, i.e. they become efficient taking into account the exogenous variable represented by number of crimes

Table 7 Efficient units (vehicle robbery)

CAMOCIM DE SAO FELIX
CUPIRA
FERREIROS
IGUARACI
JUCATI
JUPI
LAJEDO
LIMOEIRO
SALOA
SANHARO
SIRINHAEM
VITORIA DE SANTO ANTAO

The units in bold are conditionally efficient, i.e. they become efficient taking into account the exogenous variable represented by number of crimes

assessment. There is a common trade-off in the Productivity and Efficiency literature between the efficiency and the quality of the results (see e.g. Nuti et al. 2011 and Daraio et al. 2019b). Nevertheless, the issue of effectiveness is somehow neglected in the assessments of the decision unit's technical efficiency for a lack of consideration on the subjective preference and value judgments of one or many policy makers over specific rather usual production configurations.

For this assessment, we opt for a comparative analysis on the prospects from the conditional frontier evaluation associated to the effectiveness of the Pernambuco Municipalities. The effectiveness score is obtained by the ratio from the difference in the current and prior occurrences of violent crimes to the occurrences in the prior year. An effective municipality has effectiveness equal or greater than 0.12 in absolute (modulus) terms, which implies that the municipality experienced a reduction in at least 12% the incidence of violent life-related offenses. It is important to notice the non-deterministic nature in this proxy. Police departments cannot reduce crime directly. They can increase clear up rates, patrols and the operating performance, which contributes to the decrease in criminality in addition to other socio-economic determinants.

Nevertheless, the deficiency of more precise objectives by the general administration compels us to consider this approach as a close proxy for the effectiveness of the police units. Table 8 presents the effectiveness intervals and correlations. Table 9 summarizes this information presenting all the 46 effective units (31.72%) ranked according to how many times they are above the 12% goal and their relative inefficiency scores for each police production dimension.

The only effective municipality completely efficient in all three production dimensions is Camocim de São Felix, reducing 1.293 times the goal (15.38% reduction). Seven effective cities are efficient in solving violent crimes (Jucati, Jataúba, Canhotinho, Saloá, lagoa do Carro, Camocim de São Felix and Águas Belas), and only three effective municipalities (Agrestina, Bom Conselho and Camocim de São Felix) are efficient in solving Street Mugging and Vehicle Robberies. This represents only 6.21% of the total 145 municipalities (and 19.56% of the effective units) which are effective and present some level of efficiency in the production process. Most of the effective municipalities are not efficient, and many of the efficient municipalities are not effective. Table 8 presents some statistics on the

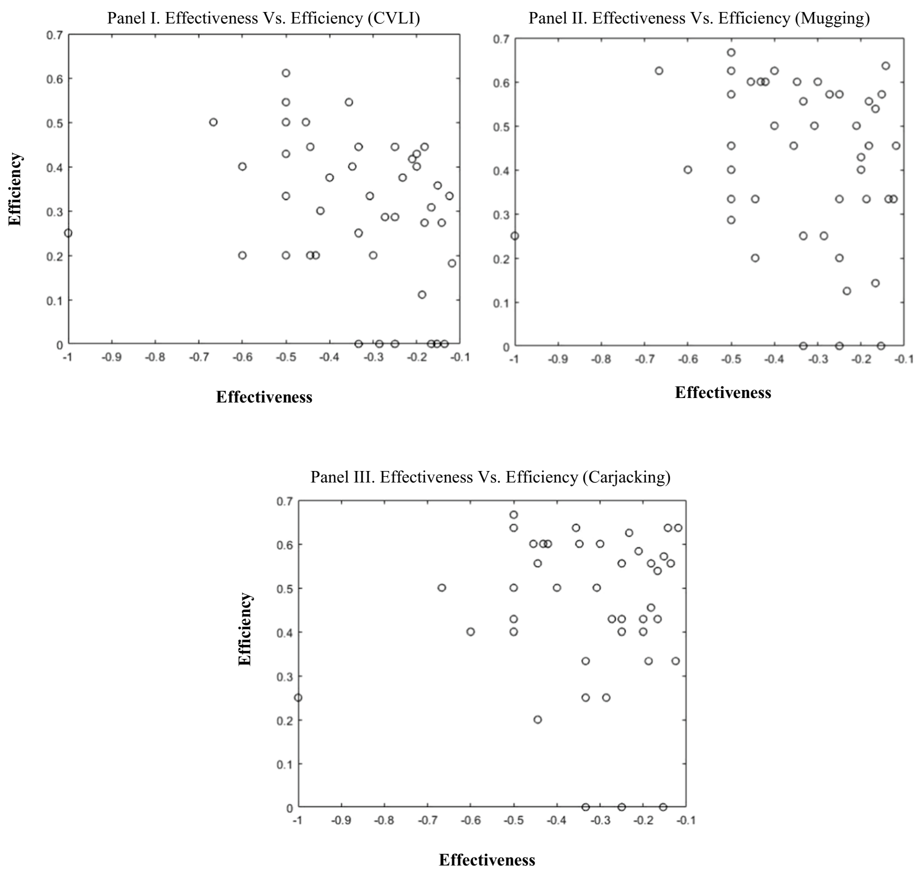

Fig. 5 Scattered associations between efficiency and effectiveness. Panel I. Effectiveness versus efficiency (CVLI) Panel II. Effectiveness versus efficiency (mugging). Panel III. Effectiveness versus efficiency (carjacking). *Note*: Effectiveness is measured by how much the target has been reached by the units. In this case, Nazaré da Mata is the reference (see Table 9), which reached the 12% target required (0.1109 to be precise). Efficiency is referred to the efficiency scores of the models with Police Officers as inputs, and respectively Violent Crimes—CVLI (Panel I), Street Mugging (Panel II),and Motor Vehicle Robberies (Carjacking, Panel III) as outputs

Table 8 Percentage distribution and correlation statistics by effectiveness intervals

Relative effectiveness		Cross-correlation statistics		
Interval	% Units	$\mu = 0.344$; sd $= 0.7637$; $\sum s^{-}(\text{iii}) = 754.28$ $h^{+} = 4.00$		
		CVLI	Street mugging	Vehicle robberies
< 1.00	–	–	–	–
−0.50– −1.00	7.76%	−0.31172	−0.11271	0.03356
−0.12– −0.49	24.14%			
0.00– −0.11	8.96%	0.00602	0.04559	0.02026
0.01–0.50	28.96%			
0.51–1.00	15.86%			
> 1.00	14.48%			

Table 9 Trade-offs between efficiency measures and effectiveness

DMU	Effectiveness	CVLI inefficiency	Street mugging inefficiency	Vehicle robberies inefficiency
CUMARU	8.403	0.25	0.25	0.25
MOREILANDIA	5.602	0.5	0.625	0.5
LAGOA DO OURO	5.042	0.2	0.4	0.4
TEREZINHA	5.042	0.4	0.4	0.4
AGUA PRETA	4.202	0.2	0.4	0.4
CALCADO	4.202	0.333333	0.333333	0.5
JATOBA	4.202	0.428571	0.571429	0.428571
JOAQUIM NABUCO	4.202	0.428571	0.285714	0.428571
QUIPAPA	4.202	0.5	0.454545	0.636364
MIRANDIBA	4.202	0.545455	0.666667	0.666667
OURICURI	4.202	0.611111	0.625	0.5
CUSTODIA	3.820	0.5	0.6	0.6
ITAQUITINGA	3.735	0.2	0.2	0.2
TABIRA	3.735	0.444444	0.333333	0.555556
CATENDE	3.629	0.2	0.6	0.6
ARACOIABA	3.538	0.3	0.6	0.6
PETROLANDIA	3.361	0.375	0.5	0.5
SANTA CRUZ	3.361	0.375	0.625	0.5
BREJO DA MADRE DE DEUS	2.988	0.545455	0.454545	0.636364
TAMANDARE	2.923	0.4	0.6	0.6
JUCATI	2.801	0	0.555556	0.333333
CORRENTES	2.801	0.25	0.25	0.25
AGRESTINA	2.801	0.444444	0	0
IPUBI	2.586	0.333333	0.5	0.5
AMARAJI	2.521	0.2	0.6	0.6
JATAUBA	2.401	0	0.25	0.25
JAQUEIRA	2.292	0.285714	0.571429	0.428571
CANHOTINHO	2.101	0	0.333333	0.555556
SALOA	2.101	0	0.2	0.4
RIACHO DAS ALMAS	2.101	0.285714	0.571429	0.428571
BOM CONSELHO	2.101	0.444444	0	0
PALMARES	1.954	0.375	0.125	0.625
ALIANCA	1.769	0.416667	0.5	0.583333
ANGELIM	1.681	0.4	0.4	0.4
BELEM DE MARIA	1.681	0.428571	0.428571	0.428571
SAO CAETANO	1.576	0.111111	0.333333	0.333333
FLORESTA	1.528	0.272727	0.454545	0.454545
SERTANIA	1.528	0.444444	0.555556	0.555556
LAGOA DO CARRO	1.401	0	0.538462	0.538462
ARARIPINA	1.401	0.307692	0.142857	0.428571
CAMOCIM DE SAO FELIX	1.293	0	0	0
TORITAMA	1.279	0.357143	0.571429	0.571429
JOAO ALFREDO	1.200	0.272727	0.636364	0.636364

Table 9 (continued)

DMU	Effectiveness	CVLI inefficiency	Street mugging inefficiency	Vehicle robberies inefficiency
AGUAS BELAS	1.146	0	0.333333	0.555556
CORTES	1.050	0.333333	0.333333	0.333333
NAZARE DA MATA	1.000	0.181818	0.454545	0.636364

The table reports the list of effective units (first column, i.e. those units that have reduced at least 12% of violent crimes), their effectiveness score (second column, i.e. the how much they reached the target of the 12% reduction) the efficiency scores calculated using Police Personal as inputs and using as outputs the CVLI clear-ups (third column), Street Mugging clear-ups (forth column) and Vehicle Robberies clear-ups (fifth column) conditional to the respective exogenous crimes

association between the two concepts. The estimates for the effective units (the intervals from -0.12 to -1.00) can be visualized in the scatter plots of Fig. 5. The negative correlation and fitting (Effectiveness vs. Efficiency Association) in the CVLI dimension suggests that higher effectiveness would be at the expense of lower efficiency. The same, but to a lower degree, can be observed in relation to the Street Mugging crime dimension.

The average 0.344 effectiveness in this assessment (Table 8) means an average municipality experienced about 34% increase in the occurrences of violent crimes from one year to the other, instead of reducing. The CVLI occurrences for 2016 reach the number 2444, increasing up to 2905 in 2017. In order to become effective, the state of Pernambuco should have reduced about 754.28 violent crimes in the specific year. The highest increase belongs to the municipality Tacaratu, having increased 400% the number of violent crimes, from 1 to 5 (lower level of effectiveness). The lack of association between the efficient units with the projected effectiveness suggests the inexistence of relevant compensations of high efficiency offsetting poor levels of effectiveness.

6 Conclusion

For the past decades of Operations Research expansion, in special regarding the Productivity and Efficiency Analysis frontier models, crime data has been used as an input for or as an output from the production technology of police departments, courts and public administrations all over the world. In this paper, we have discussed the controversial use of crime data in the assessment of the technical efficiency of decision-making units. We claim that for a given police unit, it is not possible to control social, demographic and economic determinants of criminal behaviour in the short run so that crime can be reduced as a production resource. Likewise, it is not possible to measure the number of crimes that have been prevented. The coherent output for public safety available is the number of crimes that have been solved. Therefore, it is argued in this work that efficiency analysis of police units should not rely on an investigation of crime as a discretionary input or output in the production set.

In addition to the discussion on the crime modelling in the efficiency analysis, the recent advances in the non-parametric conditional frontier models, and recent advances in the fast and efficient computation for directional measures of the technical efficiency, provide the most promissory avenue for the inclusion of criminal data as an exogenous-determined factor in efficiency and effectiveness assessments. We provide preliminary results that may encourage the extension of this approach to further analysis, based on robust conditional measures and on new datasets. These further extensions and assessments may be of particular importance for Pernambuco and many other Latin American regions facing similar levels of criminality. This kind of benchmarking analyses may be helpful to identify units as references for good practices, saving potentials and eventual consequences for public management, planning, and reward.

While efficiency does not come at the expense of lower effectiveness, additional assessments suggest effectiveness comes at the cost of reducing technical efficiency. A negative relation for high effective units (in reducing 12% or more violent crimes) as a function of inefficiencies can be inferred from the data analysis, suggesting a compensation effect, i.e. municipalities offsetting poor efficiency with high effectiveness in the CVLI dimension. This result requires a deeper interpretation of violent crimes, mostly homicides, which goes beyond the scope of this work. Nevertheless, from this potential relationship, and the information provided, one can have substantial arguments to speculate the nature of the findings, meaning (1) solving more severe violent crimes have a greater potential to prevent additional violent crimes, thus it is not necessary to be efficient for clear-up rates; (2) Policing have little to do with effectiveness.

In other words, the second hypothesis would say that police might only adequately control the efficiency in the allocation of resources to solve homicide occurrences in lieu of reducing the incidence of this exogenous factor. Due to the non-deterministic nature of the defined policy, it is fair to assume the second hypothesis as the most reliable. Assuming police cannot directly increase this specific effectiveness does not mean they cannot pursue other measures for effectiveness and control. In fact, better policies and objectives must be defined so that fairer assessments of the crime as an exogenous factor can be designed.

Lastly, we propose some interesting extensions of this work, including also the addition of more detailed data about the production process. First, the development of a comprehensive analysis based on robust conditional efficiency measures (of order-m and order-alpha) to assess the robustness of the preliminary results reported in this paper. Second, the proposal for a reallocation model given the impossibility to reduce (dismiss) public servants from their duties. Third, the extension of this analysis computing Malmquist Indices to investigate changes over time. Finally, the development of a model including policy preferences in the directional efficiency measures a priori. All these analyses are left to further research.

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Appendix: Results

See Table 10.

Table 10 Results from the directional efficiency analysis

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
AFOGADOS DA INGAZEIRA	9	11	11	0.52941	0.64706	0.64706
AFRANIO	2	3	2	0.33333	0.50000	0.33333
AGRESTINA	4	5	3	0.44444	0.55556	0.33333
AGUA PRETA	1	2	2	0.20000	0.40000	0.40000
AGUAS BELAS	0	3	5	0.00000	0.33333	0.55556
ALAGOINHA	2	2	3	0.33333	0.33333	0.50000
ALIANCA	5	6	7	0.41667	0.50000	0.58333
ALTINHO	1	0	2	0.20000	0.00000	0.40000
AMARAJI	2	6	6	0.20000	0.60000	0.60000
ANGELIM	2	2	2	0.40000	0.40000	0.40000
ARACOIABA	3	6	6	0.30000	0.60000	0.60000
ARARIPINA	4	7	7	0.30769	0.53846	0.53846
ARCOVERDE	6	10	11	0.35294	0.58824	0.64706
BARRA DE GUABIRABA	1	3	2	0.16667	0.50000	0.33333
BARREIROS	0	4	6	0.00000	0.40000	0.60000
BELEM DE MARIA	3	3	3	0.42857	0.42857	0.42857
BELEM DE SAO FRANCISCO	6	7	7	0.54545	0.63636	0.63636
BELO JARDIM	0	0	6	0.00000	0.00000	0.42857
BETANIA	3	4	3	0.42857	0.57143	0.42857
BEZERROS	6	6	5	0.37500	0.37500	0.31250
BODOCO	5	6	6	0.50000	0.60000	0.60000
BOM CONSELHO	4	3	5	0.44444	0.33333	0.55556
BOM JARDIM	1	3	5	0.11111	0.33333	0.55556
BONITO	0	2	4	0.00000	0.25000	0.50000
BREJAO	1	2	2	0.20000	0.40000	0.40000

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
BREJO DA MADRE DE DEUS	6	5	7	0.54545	0.45455	0.63636
BUIQUE	3	3	5	0.33333	0.33333	0.55556
CABO DE SANTO AGOSTINHO	0	24	21	0.00000	0.50000	0.43750
CABO DE SANTO AGOSTINHO*	**0**	**0**	**21**	**0.00000**	**0.00000**	**0.43750**
CABROBO	8	3	11	0.47059	0.17647	0.64706
CAETES	1	2	2	0.20000	0.40000	0.40000
CALCADO	2	2	3	0.33333	0.33333	0.50000
CAMARAGIBE	0	24	24	0.00000	0.54545	0.54545
CAMOCIM DE SAO FELIX	0	0	0	0.00000	0.00000	0.00000
CAMUTANGA	2	3	3	0.33333	0.50000	0.50000
CANHOTINHO	0	1	2	0.00000	0.20000	0.40000
CAPOEIRAS	1	2	2	0.20000	0.40000	0.40000
CARNAIBA	3	4	3	0.42857	0.57143	0.42857
CARPINA	6	0	11	0.30000	0.00000	0.55000
CASINHAS	0	1	1	0.00000	0.25000	0.25000
CATENDE	2	6	6	0.20000	0.60000	0.60000
CHA DE ALEGRIA	4	4	4	0.50000	0.50000	0.50000
CHA GRANDE	2	2	3	0.33333	0.33333	0.50000
CONDADO	2	3	3	0.28571	0.42857	0.42857
CORRENTES	1	1	1	0.25000	0.25000	0.25000
CORTES	2	2	2	0.33333	0.33333	0.33333
CUMARU	1	1	1	0.25000	0.25000	0.25000
CUPIRA	0	3	4	0.00000	0.33333	0.44444
CUPIRA*	**0**	**3**	**0**	**0.00000**	**0.33333**	**0.00000**

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
CUSTODIA	5	6	6	0.50000	0.60000	0.60000
ESCADA	1	11	9	0.05882	0.64706	0.52941
EXU	3	6	6	0.30000	0.60000	0.60000
FEIRA NOVA	3	2	4	0.37500	0.25000	0.50000
FERREIROS	1	1	0	0.25000	0.25000	0.00000
FLORES	6	6	6	0.60000	0.60000	0.60000
FLORESTA	3	5	5	0.27273	0.45455	0.45455
GAMELEIRA	2	4	5	0.22222	0.44444	0.55556
GLORIA DO GOITA	5	5	6	0.50000	0.50000	0.60000
GLORIA DO GOITA*	**5**	**4**	**6**	**0.50000**	**0.40000**	**0.60000**
GOIANA	0	8	13	0.00000	0.36364	0.59091
GOIANA*	**0**	**8**	**11**	**0.00000**	**0.36364**	**0.50000**
GRAVATA	10	6	14	0.50000	0.30000	0.70000
IATI	0	2	2	0.00000	0.40000	0.40000
IBIMIRIM	4	6	5	0.44444	0.66667	0.55556
IGUARACI	3	4	3	0.42857	0.57143	0.42857
IGUARACI*	**3**	**4**	**0**	**0.42857**	**0.57143**	**0.00000**
IPUBI	2	3	3	0.33333	0.50000	0.50000
ITAIBA	3	4	3	0.42857	0.57143	0.42857
ITAMBE	3	6	6	0.25000	0.50000	0.50000
ITAPISSUMA	8	10	10	0.50000	0.62500	0.62500
ITAQUITINGA	1	1	1	0.20000	0.20000	0.20000
JAQUEIRA	2	4	3	0.28571	0.57143	0.42857
JATAUBA	0	1	1	0.00000	0.25000	0.25000

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
JATOBA	3	4	3	0.42857	0.57143	0.42857
JOAO ALFREDO	4	7	7	0.36364	0.63636	0.63636
JOAO ALFREDO*	**3**	**7**	**7**	**0.27273**	**0.63636**	**0.63636**
JOAQUIM NABUCO	3	2	3	0.42857	0.28571	0.42857
JUCATI	0	0	0	0.00000	0.00000	0.00000
JUPI	1	0	0	0.16667	0.00000	0.00000
LAGOA DO CARRO	0	1	3	0.00000	0.14286	0.42857
LAGOA DO ITAENGA	1	2	2	0.12500	0.25000	0.25000
LAGOA DO OURO	1	2	2	0.20000	0.40000	0.40000
LAGOA GRANDE	1	3	2	0.16667	0.50000	0.33333
LAJEDO	3	0	0	0.27273	0.00000	0.00000
LIMOEIRO	5	1	4	0.33333	0.06667	0.26667
LIMOEIRO*	**5**	**0**	**0**	**0.33333**	**0.00000**	**0.00000**
MACAPARANA	1	4	4	0.12500	0.50000	0.50000
MACHADOS	1	2	2	0.16667	0.33333	0.33333
MIRANDIBA	6	5	7	0.54545	0.45455	0.63636
MOREILANDIA	4	5	4	0.50000	0.62500	0.50000
MORENO	9	10	12	0.50000	0.55556	0.66667
MORENO*	**9**	**0**	**12**	**0.50000**	**0.00000**	**0.66667**
NAZARE DA MATA	3	5	7	0.27273	0.45455	0.63636
NAZARE DA MATA*	**2**	**5**	**7**	**0.18182**	**0.45455**	**0.63636**
OROBO	1	2	2	0.20000	0.40000	0.40000
OROCO	2	3	3	0.33333	0.50000	0.50000
OURICURI	11	12	12	0.61111	0.66667	0.66667

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
PALMARES	6	2	10	0.37500	0.12500	0.62500
PALMEIRINA	2	2	2	0.40000	0.40000	0.40000
PANELAS	1	4	4	0.12500	0.50000	0.50000
PARANATAMA	0	1	1	0.00000	0.25000	0.25000
PARNAMIRIM	5	6	6	0.50000	0.60000	0.60000
PASSIRA	2	2	3	0.33333	0.33333	0.50000
PAUDALHO	9	4	12	0.50000	0.22222	0.66667
PESQUEIRA	5	0	7	0.35714	0.00000	0.50000
PETROLANDIA	3	4	4	0.37500	0.50000	0.50000
POMBOS	5	4	6	0.50000	0.40000	0.60000
PRIMAVERA	2	3	3	0.33333	0.50000	0.50000
QUIPAPA	4	5	4	0.50000	0.62500	0.50000
RIACHO DAS ALMAS	2	4	3	0.28571	0.57143	0.42857
RIBEIRAO	3	6	6	0.25000	0.50000	0.50000
RIO FORMOSO	2	6	6	0.20000	0.60000	0.60000
SAIRE	3	2	3	0.42857	0.28571	0.42857
SALGUEIRO	13	15	14	0.59091	0.68182	0.63636
SALOA	0	0	0	0.00000	0.00000	0.00000
SANHARO	1	2	2	0.16667	0.33333	0.33333
SANHARO*	**1**	**2**	**0**	**0.16667**	**0.33333**	**0.00000**
SANTA CRUZ	3	5	4	0.37500	0.62500	0.50000
SANTA CRUZ DO CAPIBARIBE	9	5	13	0.47368	0.26316	0.68421
SANTA MARIA DA BOA VISTA	6	9	8	0.42857	0.64286	0.57143
SANTA TEREZINHA	1	2	2	0.20000	0.40000	0.40000

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
SAO BENEDITO DO SUL	0	1	1	0.00000	0.25000	0.25000
SAO BENTO DO UNA	4	8	8	0.33333	0.66667	0.66667
SAO BENTO DO UNA*	**4**	**8**	**6**	**0.33333**	**0.66667**	**0.50000**
SAO CAETANO	1	3	3	0.11111	0.33333	0.33333
SAO JOAO	1	1	1	0.20000	0.20000	0.20000
SAO JOAQUIM DO MONTE	0	1	1	0.00000	0.20000	0.20000
SAO JOSE DA COROA GRANDE	1	5	5	0.11111	0.55556	0.55556
SAO JOSE DO BELMONTE	6	6	6	0.60000	0.60000	0.60000
SAO JOSE DO EGITO	6	5	7	0.54545	0.45455	0.63636
SAO LOURENCO DA MATA	12	6	14	0.60000	0.30000	0.70000
SAO LOURENCO DA MATA*	**12**	**0**	**14**	**0.60000**	**0.00000**	**0.70000**
SAO VICENTE FERRER	2	1	3	0.28571	0.14286	0.42857
SERRA TALHADA	16	13	14	0.66667	0.54167	0.58333
SERRITA	4	5	5	0.44444	0.55556	0.55556
SERTANIA	4	5	5	0.44444	0.55556	0.55556
SIRINHAEM	2	4	0	0.20000	0.40000	0.00000
SURUBIM	7	9	10	0.41176	0.52941	0.58824
TABIRA	4	3	5	0.44444	0.33333	0.55556
TACAIMBO	2	2	3	0.33333	0.33333	0.50000
TACARATU	2	3	3	0.33333	0.50000	0.50000
TAMANDARE	4	6	6	0.40000	0.60000	0.60000
TAQUARITINGA DO NORTE	0	4	3	0.00000	0.57143	0.42857
TEREZINHA	2	2	2	0.40000	0.40000	0.40000
TERRA NOVA	2	3	3	0.33333	0.50000	0.50000

Table 10 (continued)

DMU	FDH Directional distance measures					
	CVLI_Ineff.	Trans_Ineff.	Veic_Ineff.	CVLI_Rel.Ineff.	Trans_Rel.Ineff.	Veic_Rel.Ineff.
TIMBAUBA	6	8	11	0.35294	0.47059	0.64706
TIMBAUBA*	**0**	**8**	**11**	**0.00000**	**0.47059**	**0.64706**
TORITAMA	6	8	8	0.42857	0.57143	0.57143
TORITAMA*	**5**	**8**	**8**	**0.35714**	**0.57143**	**0.57143**
TRACUNHAEM	4	3	5	0.44444	0.33333	0.55556
TRINDADE	4	4	3	0.44444	0.44444	0.33333
TUPANATINGA	2	4	3	0.28571	0.57143	0.42857
TUPARETAMA	2	3	3	0.33333	0.50000	0.50000
VENTUROSA	1	2	3	0.12500	0.25000	0.37500
VENTUROSA*	**0**	**2**	**3**	**0.00000**	**0.25000**	**0.37500**
VERTENTE DO LERIO	1	2	2	0.20000	0.40000	0.40000
VERTENTES	1	3	3	0.14286	0.42857	0.42857
VERTENTES*	**1**	**3**	**2**	**0.14286**	**0.42857**	**0.28571**
VICENCIA	1	5	5	0.11111	0.55556	0.55556
VITORIA DE SANTO ANTAO	0	14	13	0.00000	0.50000	0.46429
VITORIA DE SANTO ANTAO*	**0**	**14**	**0**	**0.00000**	**0.50000**	**0.00000**
XEXEU	3	4	4	0.37500	0.50000	0.50000

*Units changing efficiency considering crime exogenous. Conditional FDH evaluation in bold

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